DCU

DUBLIN CITY UNIVERSITY

SEMESTER ONE EXAMINATIONS 2013

MODULE: **(Title & Code)**	CA320 Computability & Complexity
COURSE:	BSc. in Computer Applications (SE Stream)
YEAR:	3
EXAMINERS: **(Including Telephone Nos.)**	Dr. J. Power, Dr. M. Manzke, Dr. D. Sinclair, Ext no. 5510.
TIME ALLOWED:	2 hours
INSTRUCTIONS:	Answer 4 Questions. At least 1 Question **must** be from Section A. All questions carry equal marks.

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The use of programmable or text storing calculators is expressly forbidden.
Please note that where a candidate answers more than the required number of questions, the examiner will mark all questions attempted and then select the highest scoring ones

Section A

QUESTION 1 [TOTAL MARKS: 25]

1(a) [5 Marks]

In the Haskell Programming language describe how *guards* operate with the use of example code.

1(b) [10 Marks]

Let a polynomial be represented as a list of coefficients. For example $a_3x^3+a_2x^2+a_1x^1+a_0$ is represented as the list [a_0,a_1,a_2,a_3] . Define a Haskell function `evalPoly`, with the appropriate function type, which, given a polynomial and a value for x, will calculate the value of the polynomial for that value of x.

Hint: You may find the following identity helpful:
$a_nx^n + \ldots + a_2x^2 + a_1x + a_0 = a_0 + x(a_1 + x(a_2 + x(\ldots a_n)\ldots))$

1(c) [10 Marks]

Write a function `longest`, with appropriate function type, that takes a list of lists and returns the longest list in the list (and returns [ ] if the list of lists is empty).

[End of Question 1]

QUESTION 2 [TOTAL MARKS: 25]

2(a) [5 Marks]

What are *higher order functions*?

2(b) [5 Marks]

Using the higher order functions `filter` and `map`, generate a list of all the squares of numbers from 1 to 1000 that are odd.

2(c) [15 Marks]

Write a higher order sort function in Haskell and show how it can be used to sort a list of integers in increasing and decreasing order.

[End of Question 2]

Section B

QUESTION 3 [TOTAL MARKS: 25]

3(a) **[5 Marks]**

In the study of Computability & Complexity, why do we focus on languages?

3(b) **[5 Marks]**

Describe the terms *computability* and *complexity*. How do these terms relate to one another?

3(c) **[15 Marks]**

What is a *Universal Turing Machine*? What is its significance?

[End of Question 3]

QUESTION 4 [TOTAL MARKS: 25]

4(a) **[5 Marks]**

Formally define a *context-sensitive grammar*. Show how it differs from a *context-free grammar*.

4(b) **[10 Marks]**

Write a *context-sensitive grammar* that describes the language $\{a^n b^n c^n | n \geq 1\}$.

4(c) **[10 Marks]**

Formally describe a "machine", other than a Turing machine, that can recognise a *context-sensitive language*.

[End of Question 4]

QUESTION 5 [TOTAL MARKS: 25]

5(a) **[5 Marks]**

In terms of Complexity, define NP-completeness.

5(b) **[5 Marks]**

In terms of Complexity, define the *Satisfiability* (SAT) problem. In terms of complexity classes, what is the key difference between 2-SAT and 3-SAT?

5(c) **[15 Marks]**

The Cook-Levin Theorem states that *Satisfiability* is NP-complete. Give an outline of the proof of the Cook-Levin Theorem.

[End of Question 5]

[END OF EXAM]